

Program: Diploma in Artificial Intelligence & Machine Learning	
Course Code : 5343B	Course Title: Fundamentals of Internet of Things
Semester: 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic electronic circuits	2031	Fundamentals of Electrical & Electronics Engineering	II
Knowledge in Python	3347	Python Programming Lab	III

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the fundamental concepts of the Internet of Things (IoT)	14	Understand
CO2	Explain the steps in data acquisition and storage in IoT	14	Understand
CO3	Explain the components and protocols of IoT infrastructure	15	Understand
CO4	Design IoT use cases and develop IoT applications using sensors, actuators and microcontrollers.	15	Create Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Descriptions	Duration (Hours)	Cognitive Level
CO1	Explain the fundamental concepts of Internet of Things (IoT)		
M1.01	Outline the key features of Internet of Things (IoT)	3	Understand
M1.02	Explain the physical design of IOT	3	Understand
M1.03	Explain the logical design of IOT	3	Understand
M1.04	Describe different IoT-enabling technologies.	3	Understand
M1.05	Explain different levels of IoT	2	Understand
Contents: Introduction, History, Definitions for IoT, Characteristics of IoT, Components of IoT - Sensors, Actuators, Microcontrollers, Gateways Physical design of IoT- Things in IoT, Protocols Logical design of IoT: Functional blocks, Communication models, Communication APIs IoT architectures, Applications. IOT Enabling Technologies: WSN, Cloud computing, Big Data Analytics, Communication protocols, Embedded systems IoT Levels: Levels 1 to 6			
CO2	Explain the steps for data acquisition and storing data in IoT		
M2.01	Differentiate Internet of Things and Machine-to-machine communication	3	Understand
M2.02	Explain different protocols in the IoT	3	Understand
M2.03	Explain data acquisition in the IoT	4	Understand
M2.04	Explain data storage in IoT	4	Understand

Contents:

Machine-to-Machine (M2M) communication - M2M v/s. IOT.

Different protocols in IOT- MQTT, CoAP, BLE, Li-Fi, Cellular, Zigbee, WiFi, Bluetooth

Data Acquiring and Storage in IoT: Data Generation, Data Acquisition, Data Validation, Data Categorisation for Storage, Assembly Software for the Events, Data Store, Data Centre Management, Server Management, Spatial Storage

CO3	Explain the components and protocols of IoT infrastructure.		
M3.01	Describe the Role of Sensors and Actuators in IoT	2	Understand
M3.02	Classify various sensors.	2	Understand
M3.03	Demonstrate the working of sensors.	3	Apply
M3.04	Classify various actuators	2	Understand
M3.05	Demonstrate the working of Actuators	2	Apply
M3.06	Explain different protocols in IoT	4	Understand

Contents:

Sensor & Actuators: - Role of Sensors and Actuators in IoT

Sensors - Overview, working, Analog and Digital.

Various types of sensors - Temperature, Distance, Humidity, Motion, Light and Gas. Sensor fusion

Actuators - Overview, working.

Various types of actuators - Relay, LEDs, Buzzer, DCMotor and Servo Motor.

CO4	Design IoT use cases and develop IoT applications using sensors, actuators and microcontrollers.		
M4.01	List embedded computing boards	1	Understand
M4.02	Describe the features of embedded computing boards - Arduino/Node MCU & Raspberry PI	4	Understand
M4.03	Illustrate the interfacing of basic sensors with embedded computing board - Arduino/Node MCU & Raspberry PI	5	Apply
M4.04	Develop IoT applications using Arduino/Raspberry PI , sensors, actuators and breadboards	5	Apply

Contents:

Embedded Computing Boards - Features and characteristics of Arduino UNO/Node MCU & Raspberry PI, Interfacing basic sensors with Computing boards

Programming Raspberry Pi with python (modules, packages, input/output), Interfacing sensors and actuators with Raspberry PI.
Various applications/use cases with technology integration.

Text / Reference

T/R	Book Title/Author
T1	Vijay Madisetti, Arshdeep Bahga, Internet of Things: A Hands-On Approach, Orient Blackswan
T2	Raj Kamal, Internet of Things : Architecture And Design Principles, McGraw Hill Education
R1	RMD Sundaram Shriram K Vasudevan, Abhishek S Nagarajan, Internet of Things, Wiley Publications
R2	Marco Schwartz, Internet of Things with Arduino Cookbook, Packt Publishing

Online Resources

Sl.No	Website Link
1	https://developers.google.com/machine-learning/crash-course/
2	https://studyglance.in/labprograms/mllabprograms.php
3	https://machinehack.com/